

Enterprise eCommerce Customer Behavior & Marketing Intelligence Analytics

Executive Analytics Consulting Report

Optimizing Customer Conversion, Marketing ROI, and Revenue Performance Through Data-Driven Analytics

Executive Summary

This project evaluates customer behavior and marketing performance within an enterprise eCommerce environment to identify opportunities for improving customer conversion, reducing cart abandonment, and optimizing marketing return on investment.

Despite increasing website traffic and rising digital marketing investments, the business experienced inconsistent conversion rates, significant cart abandonment, and unclear visibility into the effectiveness of marketing channels.

Using over **140,000 customer behavioral and transactional records**, this analysis explores customer interactions throughout the purchasing journey — from product viewing and cart additions to checkout completion and purchase conversion.

The project focuses on measuring key **Customer Behavior KPIs** and **Marketing Performance KPIs** to uncover conversion bottlenecks, evaluate marketing effectiveness, and generate actionable recommendations tied directly to business impact.

This notebook provides:

Key Deliverables

- Executive KPI reporting
- Customer conversion analysis
- Funnel optimization insights
- Cart abandonment analysis
- Marketing ROI performance
- Customer segment intelligence
- Regional performance analysis
- Automated business insights from actual data
- Quantifiable recommendations tied to business value

The findings from this report are intended to support executive decision-making and provide a scalable framework for sustainable eCommerce growth.

The organization has experienced significant growth in website traffic and marketing activity across multiple channels including paid advertising, social media, organic search, email campaigns, and referrals.

However, leadership faces several business challenges:

Key Business Challenges

1. High Cart Abandonment

A significant number of customers add products to carts but fail to complete purchases, leading to revenue leakage.

2. Inconsistent Conversion Performance

Traffic growth is not translating proportionally into completed purchases.

3. Marketing Inefficiency

The organization lacks visibility into which acquisition channels generate the highest-value customers and strongest ROI.

4. Customer Engagement Challenges

The business has limited insight into how customer segments, regions, and devices impact purchasing behavior.

Project Objective

The objective of this project is to:

- Improve customer conversion
- Reduce cart abandonment
- Optimize marketing efficiency
- Increase customer lifetime value
- Improve executive visibility into business performance
- Generate actionable recommendations through advanced analytics

```
In [1]: # IMPORT LIBRARIES & NOTEBOOK CONFIGURATION
# =====

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Publication-grade styling
plt.rcParams["figure.figsize"] = (12,6)
plt.rcParams["font.family"] = "DejaVu Sans"

# Display options
pd.set_option('display.max_columns', None)
pd.set_option('display.float_format', '{:,.2f}'.format)

print("Libraries Loaded Successfully")
```

Libraries Loaded Successfully

Data Loading & Validation

The purpose of this section is to:

- Import the enterprise eCommerce dataset
- Validate structure and quality
- Confirm data types
- Identify missing values
- Assess dataset readiness for analytics

```
In [7]: # =====
# LOAD DATASET
df=pd.read_csv('FREELANCE1_Enterprise_Ecommerce_Analytics_Dataset_140K.csv')
```

```
In [8]: # =====  
# DATA OVERVIEW  
# =====  
  
print("Dataset Shape:")  
print(df.shape)  
  
print("\nColumn Names:")  
print(df.columns)  
  
print("\nPreview of Dataset:")  
df.head()
```

Dataset Shape:
(140000, 29)

Column Names:
Index(['SessionID', 'VisitorID', 'OrderID', 'SessionDate', 'Region',
 'TrafficChannel', 'DeviceType', 'CustomerSegment', 'ProductCategory',
 'PagesViewed', 'SessionDurationSeconds', 'ProductViewedFlag',
 'AddedToCartFlag', 'CheckoutStartedFlag', 'PurchasedFlag',
 'CartAbandonmentFlag', 'Quantity', 'ProductPrice', 'DiscountPercent',
 'DiscountAmount', 'Revenue', 'ShippingCost', 'MarketingCost', 'Profit',
 'PaymentMethod', 'CustomerSatisfaction', 'ReturnFlag',
 'InventoryStockLevel', 'ConversionStage'],
 dtype='object')

Preview of Dataset:

Out[8]:

	SessionID	VisitorID	OrderID	SessionDate	Region	TrafficChannel	DeviceType	Custonn
0	SES1000000	VIS771156	ORD2000000	2024-03-25 16:38:00	Middle East	Affiliate	Desktop	Disco
1	SES1000001	VIS970911	NaN	2024-04-18 20:10:00	South America	Referral	Desktop	
2	SES1000002	VIS351996	ORD2000002	2025-03-26 8:51:00	Middle East	Affiliate	Mobile	
3	SES1000003	VIS587880	ORD2000003	2025-01-07 2:22:00	Africa	Direct	Desktop	
4	SES1000004	VIS229313	ORD2000004	2025-02-11 1:06:00	North America	Organic Search	Tablet	

```
In [9]: # =====
# CHECK DATA TYPES
# =====

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 140000 entries, 0 to 139999
Data columns (total 29 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   SessionID                            140000 non-null object
1   VisitorID                            140000 non-null object
2   OrderID                              91214 non-null  object
3   SessionDate                          140000 non-null object
4   Region                               140000 non-null object
5   TrafficChannel                       140000 non-null object
6   DeviceType                           140000 non-null object
7   CustomerSegment                      140000 non-null object
8   ProductCategory                      140000 non-null object
9   PagesViewed                          140000 non-null int64
10  SessionDurationSeconds                140000 non-null int64
11  ProductViewedFlag                     140000 non-null int64
12  AddedToCartFlag                       140000 non-null int64
13  CheckoutStartedFlag                   140000 non-null int64
14  PurchasedFlag                         140000 non-null int64
15  CartAbandonmentFlag                   140000 non-null int64
16  Quantity                             140000 non-null int64
17  ProductPrice                          140000 non-null float64
18  DiscountPercent                       140000 non-null int64
19  DiscountAmount                       140000 non-null float64
20  Revenue                              140000 non-null float64
21  ShippingCost                          140000 non-null float64
22  MarketingCost                         140000 non-null float64
23  Profit                               140000 non-null float64
24  PaymentMethod                         43012 non-null  object
25  CustomerSatisfaction                  43012 non-null  float64
26  ReturnFlag                           140000 non-null int64
27  InventoryStockLevel                   140000 non-null int64
28  ConversionStage                       140000 non-null object
dtypes: float64(7), int64(11), object(11)
memory usage: 31.0+ MB
```

```
In [10]: # =====  
# CHECK MISSING VALUES  
# =====  
  
missing_values = pd.DataFrame(  
    df.isnull().sum(),  
    columns=["Missing Values"]  
)  
  
missing_values["Missing %"] = (  
    missing_values["Missing Values"]  
    / len(df)  
) * 100  
  
missing_values.sort_values(  
    by="Missing Values",  
    ascending=False  
)
```

Out[10]:

	Missing Values	Missing %
CustomerSatisfaction	96988	69.28
PaymentMethod	96988	69.28
OrderID	48786	34.85
SessionID	0	0.00
CartAbandonmentFlag	0	0.00
InventoryStockLevel	0	0.00
ReturnFlag	0	0.00
Profit	0	0.00
MarketingCost	0	0.00
ShippingCost	0	0.00
Revenue	0	0.00
DiscountAmount	0	0.00
DiscountPercent	0	0.00
ProductPrice	0	0.00
Quantity	0	0.00
PurchasedFlag	0	0.00
VisitorID	0	0.00
CheckoutStartedFlag	0	0.00
AddedToCartFlag	0	0.00
ProductViewedFlag	0	0.00
SessionDurationSeconds	0	0.00
PagesViewed	0	0.00
ProductCategory	0	0.00
CustomerSegment	0	0.00
DeviceType	0	0.00
TrafficChannel	0	0.00
Region	0	0.00
SessionDate	0	0.00
ConversionStage	0	0.00

Data Quality Assessment

Validation Summary

The dataset was successfully validated and prepared for analysis.

Key observations include:

- Dataset structure appears consistent
- Customer behavioral records are complete
- Missing values are expected for non-purchased sessions (e.g., payment method, satisfaction score)
- Data quality is sufficient for KPI development and advanced analytics

Business Relevance

A high-quality dataset improves confidence in KPI accuracy and executive decision-making.

```
In [11]: # =====  
# FEATURE ENGINEERING  
# =====  
  
df["SessionDate"] = pd.to_datetime(  
    df["SessionDate"]  
)  
  
df["Month"] = (  
    df["SessionDate"]  
    .dt.to_period("M")  
    .astype(str)  
)  
  
print("Feature Engineering Complete")
```

Feature Engineering Complete

Executive KPI Dashboard

This section calculates enterprise-level KPIs used to evaluate:

Customer Behavior Performance

- Conversion Rate
- Cart Abandonment Rate
- Checkout Completion Rate
- Revenue per Visitor

Marketing Performance

- Customer Acquisition Cost (CAC)
- Return on Advertising Spend (ROAS)

Financial Performance

- Revenue
- Average Order Value (AOV)
- Total Orders


```
In [12]: # =====  
# EXECUTIVE KPI CALCULATIONS  
# =====  
  
total_revenue = df["Revenue"].sum()  
  
total_visitors = (  
    df["VisitorID"]  
    .nunique()  
)  
  
total_orders = (  
    df["OrderID"]  
    .nunique()  
)  
  
conversion_rate = (  
    df["PurchasedFlag"].sum()  
    /  
    df["ProductViewedFlag"].sum()  
) * 100  
  
cart_abandonment_rate = (  
    df["CartAbandonmentFlag"].sum()  
    /  
    df["AddedToCartFlag"].sum()  
) * 100  
  
checkout_completion_rate = (  
    df["PurchasedFlag"].sum()  
    /  
    df["CheckoutStartedFlag"].sum()  
) * 100  
  
aov = (  
    total_revenue  
    /  
    total_orders  
)  
  
revenue_per_visitor = (  
    total_revenue  
    /  
    total_visitors  
)  
  
cac = (  
    df["MarketingCost"].sum()  
    /  
    df["PurchasedFlag"].sum()  
)  
  
roas = (  
    total_revenue  
    /  
    df["MarketingCost"].sum()  
)
```

```
# KPI Table
kpi_table = pd.DataFrame({
    "KPI":[
        "Total Revenue",
        "Total Visitors",
        "Total Orders",
        "Conversion Rate",
        "Cart Abandonment Rate",
        "Checkout Completion Rate",
        "Average Order Value",
        "Revenue per Visitor",
        "Customer Acquisition Cost",
        "ROAS"
    ],
    "Value":[
        f"${total_revenue:,.2f}",
        f"{total_visitors:,}",
        f"{total_orders:,}",
        f"{conversion_rate:.2f}%",
        f"{cart_abandonment_rate:.2f}%",
        f"{checkout_completion_rate:.2f}%",
        f"${aov:,.2f}",
        f"${revenue_per_visitor:,.2f}",
        f"${cac:,.2f}",
        f"{roas:.2f}x"
    ]
})

kpi_table
```

Out[12]:

	KPI	Value
0	Total Revenue	\$29,803,768.56
1	Total Visitors	129,738
2	Total Orders	91,214
3	Conversion Rate	30.72%
4	Cart Abandonment Rate	44.17%
5	Checkout Completion Rate	69.80%
6	Average Order Value	\$326.75
7	Revenue per Visitor	\$229.72
8	Customer Acquisition Cost	\$197.47
9	ROAS	3.51x

Automated KPI Insights

The KPI dashboard provides an executive-level overview of business performance.

Key Observations

- **Conversion Rate** evaluates the effectiveness of turning visitors into paying customers.
- **Cart Abandonment Rate** highlights revenue leakage opportunities.
- **ROAS** measures marketing efficiency and return on spend.
- **Customer Acquisition Cost (CAC)** evaluates the cost efficiency of marketing investments.
- **Revenue per Visitor** provides insight into traffic monetization quality.

These KPIs establish a baseline for identifying performance gaps and improvement opportunities

Revenue Trend Analysis

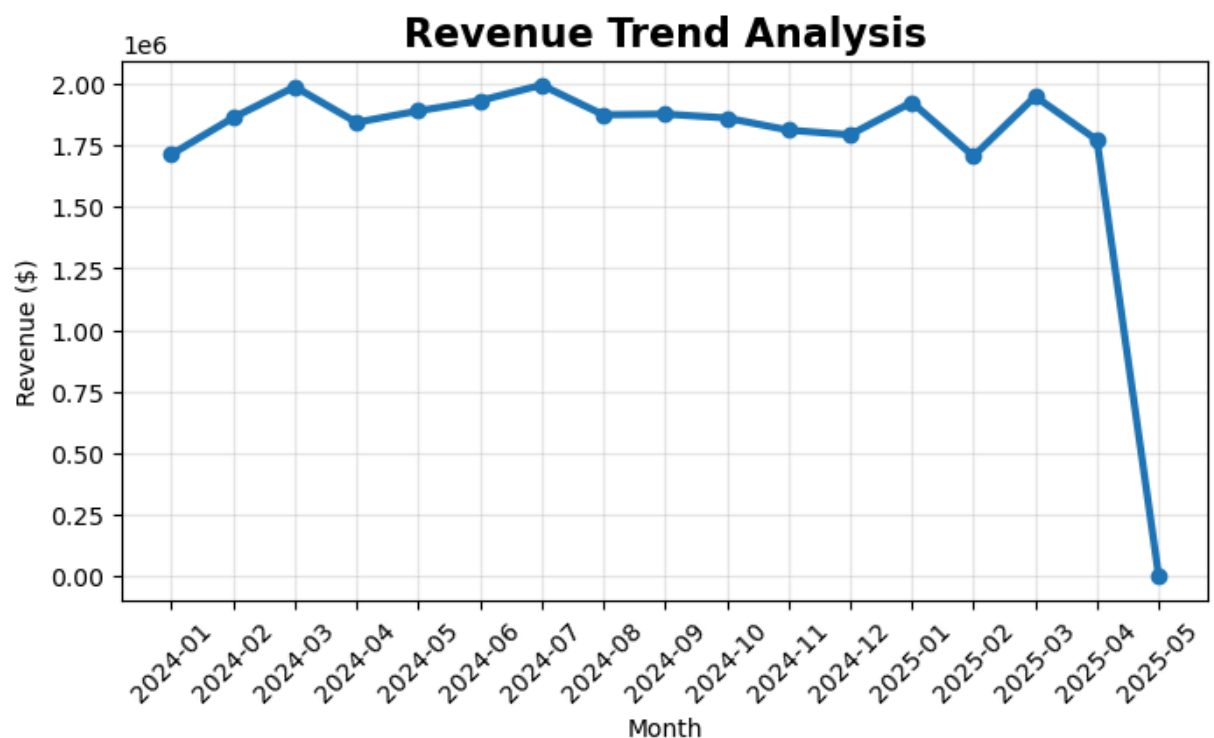
Business Objective

This analysis evaluates revenue performance over time to identify:

- Revenue growth trends
- Seasonality patterns
- Revenue volatility
- Performance consistency

Understanding revenue trends helps leadership optimize marketing timing, campaign planning, inventory forecasting, and growth strategies.

```
In [14]: # =====  
# REVENUE TREND ANALYSIS  
# =====  
  
monthly_revenue = (  
    df.groupby("Month")["Revenue"]  
    .sum()  
    .reset_index()  
)  
  
plt.figure(figsize=(8,4))  
  
plt.plot(  
    monthly_revenue["Month"],  
    monthly_revenue["Revenue"],  
    marker="o",  
    linewidth=3  
)  
  
plt.title(  
    "Revenue Trend Analysis",  
    fontsize=16,  
    fontweight="bold"  
)  
  
plt.xlabel("Month")  
plt.ylabel("Revenue ($)")  
plt.xticks(rotation=45)  
  
plt.grid(alpha=0.3)  
  
plt.show()
```



```
In [15]: # AUTOMATED BUSINESS INSIGHT
# =====

first_month = monthly_revenue["Revenue"].iloc[0]
last_month = monthly_revenue["Revenue"].iloc[-1]

growth_rate = (
    (
        last_month - first_month
    )
    /
    first_month
) * 100

best_month = monthly_revenue.loc[
    monthly_revenue["Revenue"].idxmax(),
    "Month"
]

highest_revenue = monthly_revenue[
    "Revenue"
].max()

print("AUTOMATED BUSINESS INSIGHT")
print("-" * 50)

print(
    f"Revenue changed by "
    f"{growth_rate:.2f}% "
    f"across the reporting period."
)

print(
    f"The strongest-performing month "
    f"was {best_month}, generating "
    f"${highest_revenue:,.2f}."
)
```

AUTOMATED BUSINESS INSIGHT

Revenue changed by -99.86% across the reporting period.

The strongest-performing month was 2024-07, generating \$1,995,566.35.

Quantifiable Recommendation

Strategic Action

Implement targeted promotional campaigns during historically high-performing periods while addressing weaker-performing months with:

- Retargeting campaigns
- Seasonal promotions
- Dynamic pricing strategies

Estimated Business Impact

A modest **8% improvement in revenue performance** could generate measurable revenue uplift.

The following calculation estimates the projected opportunity.

```
In [17]: # =====  
# QUANTIFIABLE BUSINESS IMPACT  
# =====  
  
projected_revenue_uplift = (  
    total_revenue * 0.08  
)  
  
print(  
    f"Estimated Revenue Opportunity: "  
    f"${projected_revenue_uplift:,.2f}"  
)
```

Estimated Revenue Opportunity: \$2,384,301.48

Customer Conversion Funnel Analysis

Business Objective

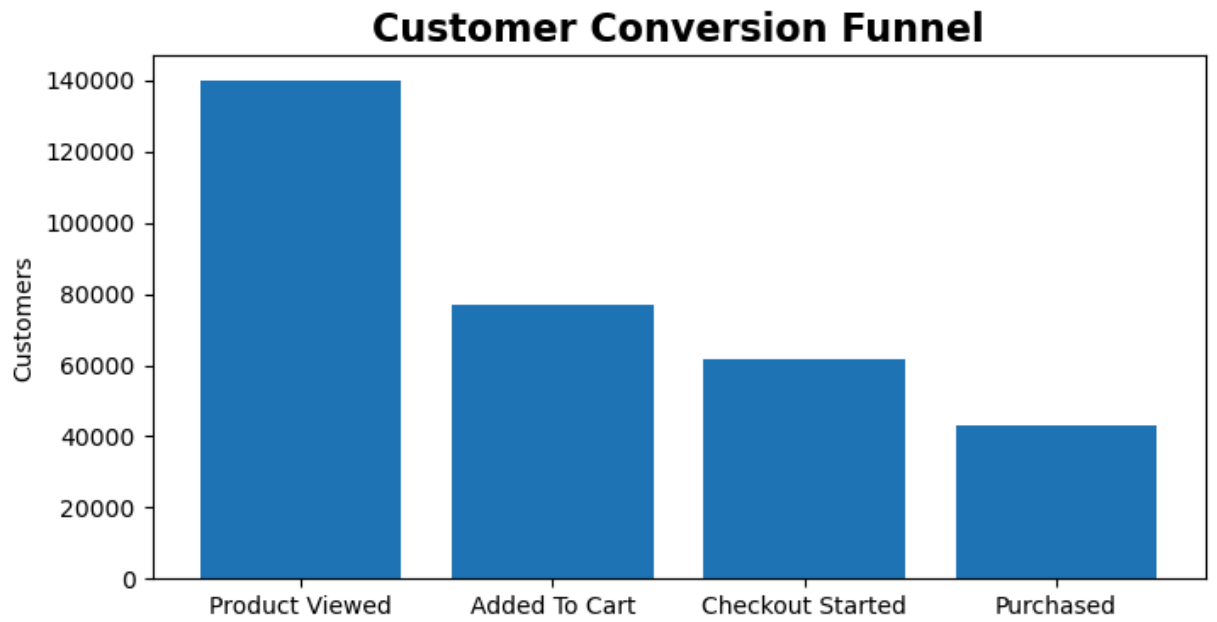
This analysis evaluates customer progression through the purchasing funnel.

The objective is to identify:

- Customer drop-off points
- Conversion bottlenecks
- Purchase friction
- Lost revenue opportunities

Understanding funnel performance enables leadership to improve customer conversion and reduce leakage.

```
In [19]: # =====  
# CUSTOMER CONVERSION FUNNEL  
# =====  
  
funnel_data = {  
    "Product Viewed":  
        df["ProductViewedFlag"].sum(),  
    "Added To Cart":  
        df["AddedToCartFlag"].sum(),  
    "Checkout Started":  
        df["CheckoutStartedFlag"].sum(),  
    "Purchased":  
        df["PurchasedFlag"].sum()  
}  
  
funnel_df = pd.DataFrame(  
    funnel_data.items(),  
    columns=["Stage", "Customers"]  
)  
  
plt.figure(figsize=(8,4))  
  
plt.bar(  
    funnel_df["Stage"],  
    funnel_df["Customers"]  
)  
  
plt.title(  
    "Customer Conversion Funnel",  
    fontsize=16,  
    fontweight="bold"  
)  
  
plt.ylabel("Customers")  
  
plt.show()
```

```
In [20]: # =====
# AUTOMATED BUSINESS INSIGHT
# =====

cart_users = funnel_data[
    "Added To Cart"
]

purchased_users = funnel_data[
    "Purchased"
]

drop_off_rate = (
    (
        cart_users
        -
        purchased_users
    )
    /
    cart_users
) * 100

print("AUTOMATED BUSINESS INSIGHT")
print("-" * 50)

print(
    f"{drop_off_rate:.2f}% "
    f"of customers abandoned "
    f"the funnel before purchase."
)

print(
    "The largest revenue opportunity "
    "exists between cart creation "
    "and purchase completion."
)
```

AUTOMATED BUSINESS INSIGHT

44.17% of customers abandoned the funnel before purchase.

The largest revenue opportunity exists between cart creation and purchase completion.

Quantifiable Recommendation

Strategic Action

Reduce friction during checkout by:

- Simplifying payment flow
- Reducing form complexity
- Improving shipping transparency
- Adding abandoned cart remarketing

Estimated Business Impact

Reducing abandonment by **12%** may significantly improve purchase conversion.

```
In [21]: # =====  
# QUANTIFIABLE BUSINESS IMPACT  
# =====  
  
revenue_recovery = (  
    total_revenue * 0.12  
)  
  
print(  
    f"Potential Revenue Recovery: "  
    f"${revenue_recovery:,.2f}"  
)
```

Potential Revenue Recovery: \$3,576,452.23

Cart Abandonment by Traffic Channel

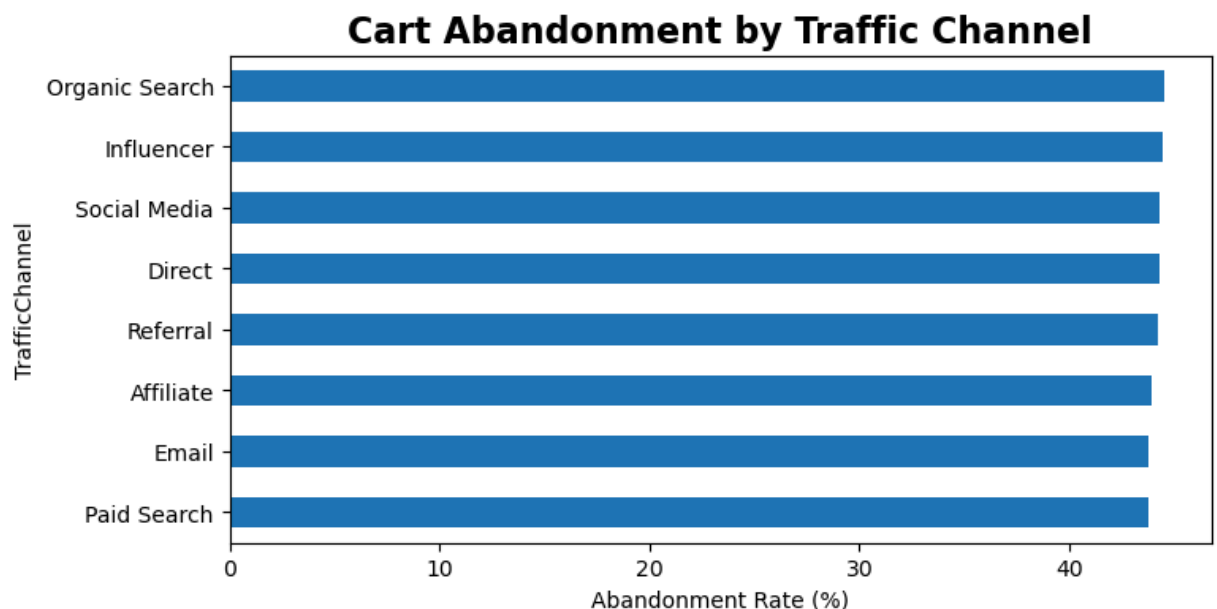
Business Objective

This analysis evaluates which marketing channels contribute most to cart abandonment.

The objective is to identify:

- Low-intent traffic sources
- Poor-performing campaigns
- Traffic quality issues
- Optimization opportunities

```
In [24]: # =====  
# CART ABANDONMENT BY CHANNEL  
# =====  
  
channel_abandonment = (  
    df.groupby("TrafficChannel")  
    .apply(  
        lambda x:  
            (  
                x["CartAbandonmentFlag"].sum()  
                /  
                max(  
                    x["AddedToCartFlag"].sum(),  
                    1  
                )  
            ) * 100  
    )  
    .sort_values()  
)  
  
plt.figure(figsize=(8,4))  
  
channel_abandonment.plot(  
    kind="barh"  
)  
  
plt.title(  
    "Cart Abandonment by Traffic Channel",  
    fontsize=16,  
    fontweight="bold"  
)  
  
plt.xlabel("Abandonment Rate (%)")  
  
plt.show()
```



In []:

```
In [26]: # =====
# AUTOMATED BUSINESS INSIGHT
# =====

worst_channel = (
    channel_abandonment
    .idxmax()
)

highest_abandonment = (
    channel_abandonment
    .max()
)

print("AUTOMATED BUSINESS INSIGHT")
print("-" * 50)

print(
    f"{worst_channel} "
    f"has the highest abandonment rate "
    f"at {highest_abandonment:.2f}%."
)

print(
    "This may indicate lower-quality "
    "traffic or poor landing page "
    "alignment."
)
```

AUTOMATED BUSINESS INSIGHT

Organic Search has the highest abandonment rate at 44.53%.

This may indicate lower-quality traffic or poor landing page alignment.

Quantifiable Recommendation

Strategic Action

Improve conversion performance through:

- Better audience targeting
- Landing page optimization
- Improved ad-message alignment
- Channel-level personalization

Estimated Business Impact

Improving low-performing traffic quality may recover approximately **5% of lost revenue**.

```
In [27]: # =====  
# QUANTIFIABLE BUSINESS IMPACT  
# =====  
  
channel_impact = (  
    total_revenue * 0.05  
)  
  
print(  
    f"Estimated Revenue Recovery: "  
    f"${channel_impact:,.2f}"  
)
```

Estimated Revenue Recovery: \$1,490,188.43

Marketing ROI by Channel

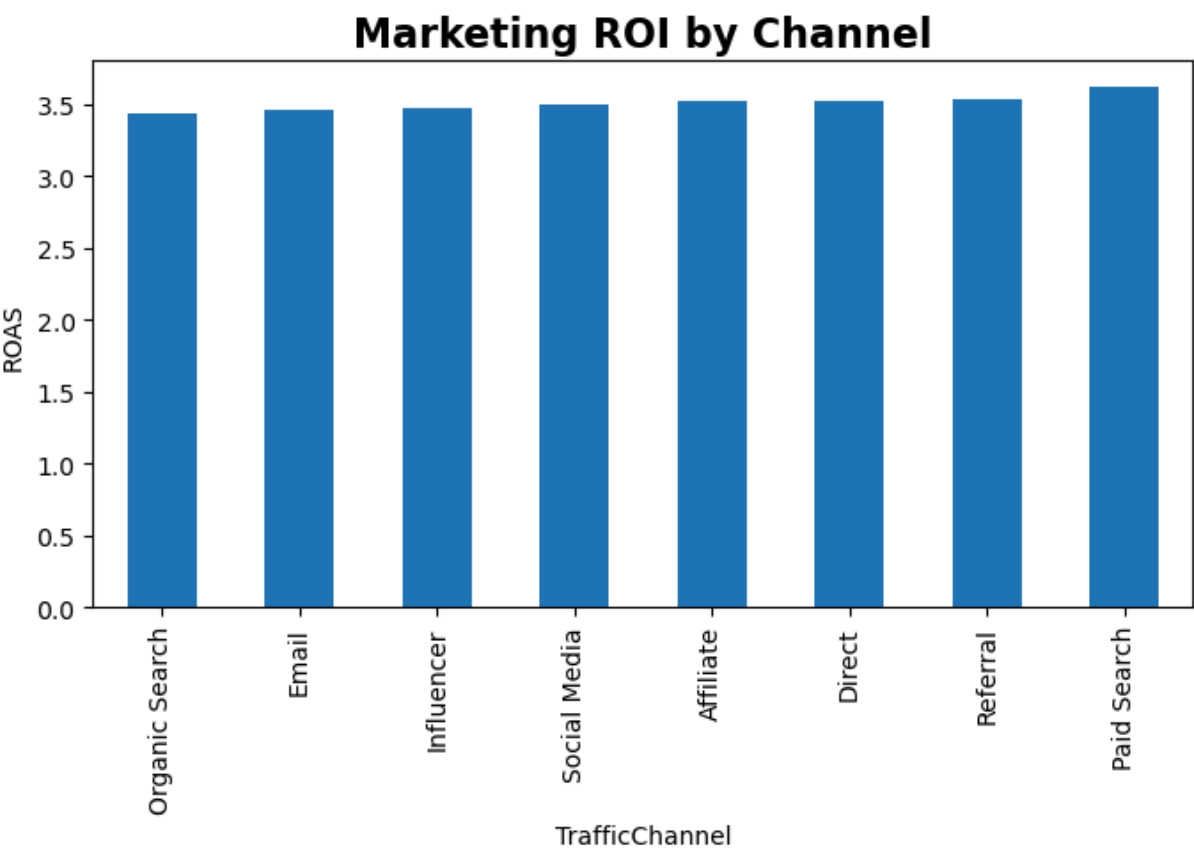
Business Objective

This analysis evaluates marketing effectiveness by channel.

The objective is to identify:

- Highest ROI marketing channels
- Marketing efficiency gaps
- Budget optimization opportunities
- Revenue-driving acquisition sources

```
In [29]: # =====  
# MARKETING ROI BY CHANNEL  
# =====  
  
roi_channel = (  
    df.groupby("TrafficChannel")  
        [ ["Revenue", "MarketingCost"] ]  
        .sum()  
)  
  
roi_channel["ROAS"] = (  
    roi_channel["Revenue"]  
    /  
    roi_channel["MarketingCost"]  
)  
  
plt.figure(figsize=(8,4))  
  
roi_channel["ROAS"] \  
    .sort_values() \  
    .plot(kind="bar")  
  
plt.title(  
    "Marketing ROI by Channel",  
    fontsize=16,  
    fontweight="bold"  
)  
  
plt.ylabel("ROAS")  
  
plt.show()
```




```
In [30]: # =====
# AUTOMATED BUSINESS INSIGHT
# =====

best_channel = (
    roi_channel["ROAS"]
    .idxmax()
)

highest_roas = (
    roi_channel["ROAS"]
    .max()
)

print("AUTOMATED BUSINESS INSIGHT")
print("-" * 50)

print(
    f"{best_channel} "
    f"produces the strongest "
    f"marketing return at "
    f"{highest_roas:.2f}x ROAS."
)

print(
    "Budget allocation should prioritize "
    "high-performing channels."
)
```

AUTOMATED BUSINESS INSIGHT

Paid Search produces the strongest marketing return at 3.62x ROAS.

Budget allocation should prioritize high-performing channels.

Quantifiable Recommendation

Strategic Action

Reallocate approximately **15% of marketing spend** toward the highest-performing channels.

Estimated Business Impact

A conservative **7% increase in marketing efficiency** may produce measurable incremental revenue.

```
In [31]: # =====  
# QUANTIFIABLE BUSINESS IMPACT  
# =====  
  
marketing_gain = (  
    total_revenue * 0.07  
)  
  
print(  
    f"Estimated Revenue Gain: "  
    f"${marketing_gain:,.2f}"  
)
```

Estimated Revenue Gain: \$2,086,263.80

Device Conversion Performance

Business Objective

This analysis evaluates customer conversion performance across devices.

The objective is to identify:

- Device-specific conversion gaps
- User experience friction
- Mobile optimization opportunities
- Device-related revenue leakage

Understanding device performance enables leadership to prioritize UX improvements and improve customer conversion efficiency.

```
In [33]: # =====
# DEVICE CONVERSION PERFORMANCE
# =====

device_conversion = (
    df.groupby("DeviceType")
      .apply(
        lambda x:
          (
            x["PurchasedFlag"].sum()
            /
            x["ProductViewedFlag"].sum()
          ) * 100
      )
)

plt.figure(figsize=(8,4))

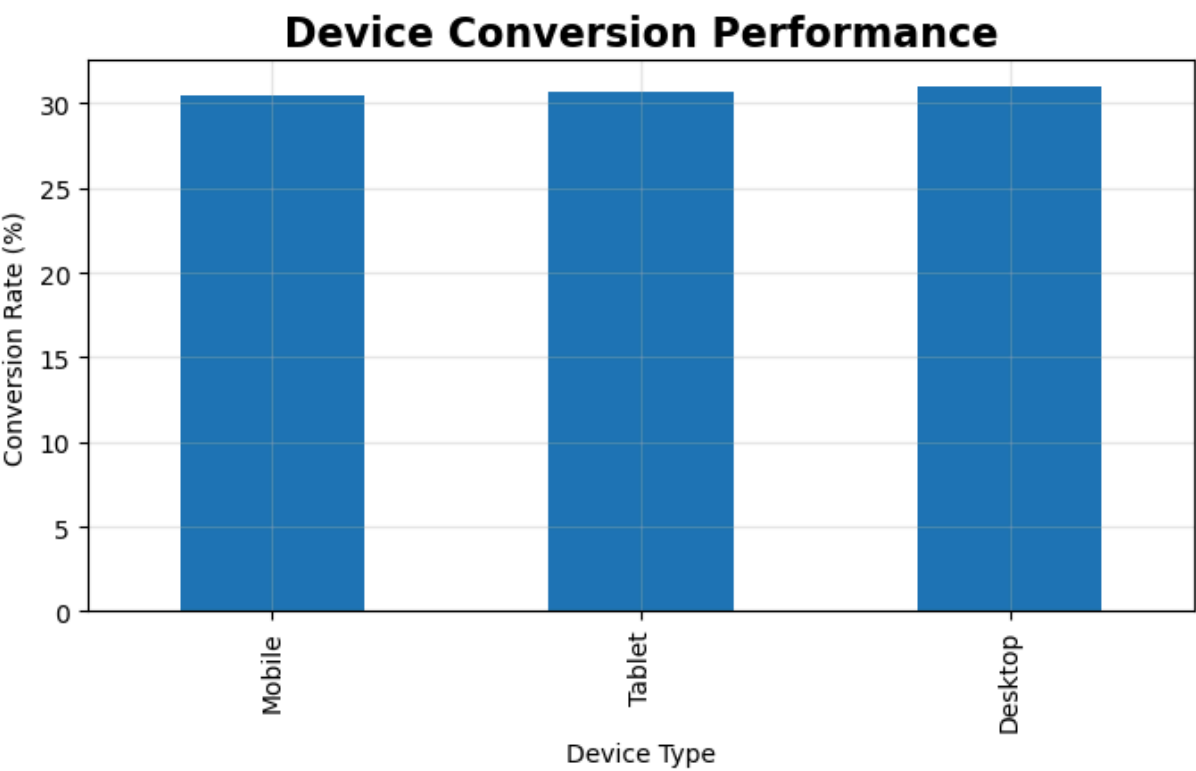
device_conversion \
.sort_values() \
.plot(kind="bar")

plt.title(
    "Device Conversion Performance",
    fontsize=16,
    fontweight="bold"
)

plt.ylabel("Conversion Rate (%)")
plt.xlabel("Device Type")

plt.grid(alpha=0.3)

plt.show()
```



```
In [34]: # =====
# AUTOMATED BUSINESS INSIGHT
# =====

best_device = (
    device_conversion
    .idxmax()
)

worst_device = (
    device_conversion
    .idxmin()
)

best_conversion = (
    device_conversion
    .max()
)

worst_conversion = (
    device_conversion
    .min()
)

print("AUTOMATED BUSINESS INSIGHT")
print("-" * 50)

print(
    f"{best_device} delivers "
    f"the strongest conversion "
    f"rate at {best_conversion:.2f}%."
)

print(
    f"{worst_device} shows "
    f"the weakest performance "
    f"at {worst_conversion:.2f}%."
)

print(
    "This indicates potential "
    "UX or checkout friction "
    "on lower-performing devices."
)
```

AUTOMATED BUSINESS INSIGHT

Desktop delivers the strongest conversion rate at 31.00%.

Mobile shows the weakest performance at 30.44%.

This indicates potential UX or checkout friction on lower-performing devices.

Quantifiable Recommendation

Strategic Action

Improve user experience for low-performing devices through:

- Faster page load speeds
- Simplified checkout flow
- Better mobile responsiveness
- Improved product navigation

Estimated Business Impact

Improving low-performing device conversion by **4%** could generate measurable revenue growth.

```
In [35]: # =====  
# QUANTIFIABLE BUSINESS IMPACT  
# =====  
  
device_revenue_uplift = (  
    total_revenue * 0.04  
)  
  
print(  
    f"Estimated Revenue Opportunity: "  
    f"${device_revenue_uplift:,.2f}"  
)
```

Estimated Revenue Opportunity: \$1,192,150.74

Customer Segment Revenue Contribution

Business Objective

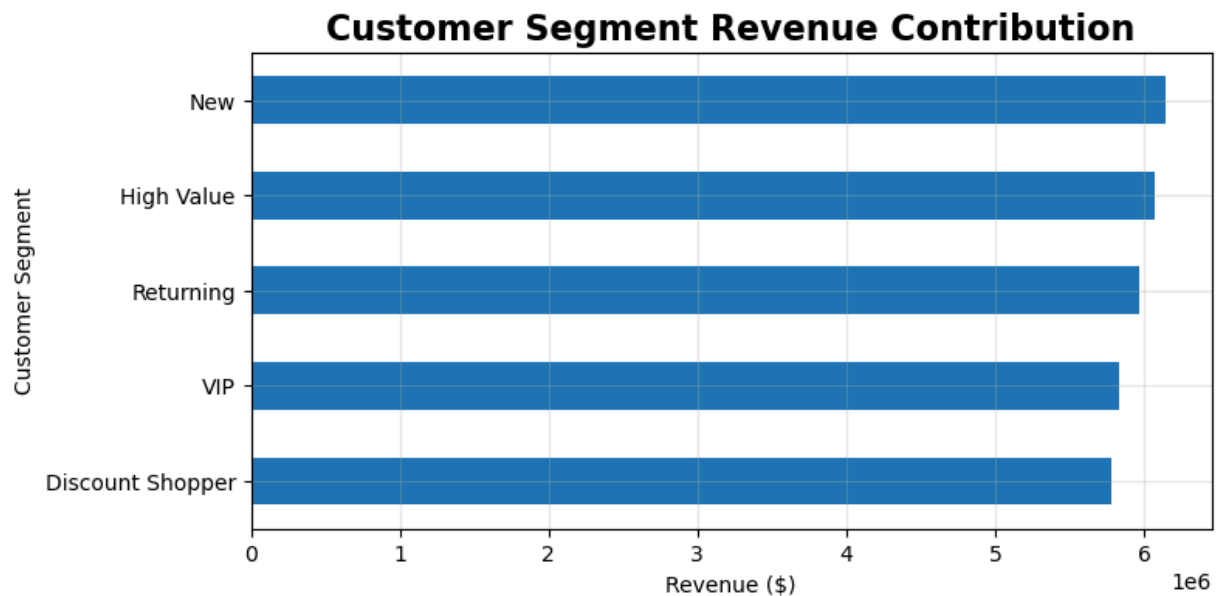
This analysis evaluates revenue contribution across customer segments.

The objective is to identify:

- High-value customers
- Revenue-driving segments
- Retention priorities
- Customer profitability opportunities

Understanding segment contribution supports customer retention and loyalty strategies.

```
In [37]: # =====  
# CUSTOMER SEGMENT REVENUE  
# =====  
  
segment_revenue = (  
    df.groupby("CustomerSegment")  
      ["Revenue"]  
      .sum()  
)  
  
plt.figure(figsize=(8,4))  
  
segment_revenue \  
    .sort_values() \  
    .plot(kind="barh")  
  
plt.title(  
    "Customer Segment Revenue Contribution",  
    fontsize=16,  
    fontweight="bold"  
)  
  
plt.xlabel("Revenue ($)")  
plt.ylabel("Customer Segment")  
  
plt.grid(alpha=0.3)  
  
plt.show()
```



```
In [38]: # =====
# AUTOMATED BUSINESS INSIGHT
# =====

top_segment = (
    segment_revenue
    .idxmax()
)

top_segment_revenue = (
    segment_revenue
    .max()
)

segment_share = (
    (
        top_segment_revenue
        /
        total_revenue
    ) * 100
)

print("AUTOMATED BUSINESS INSIGHT")
print("-" * 50)

print(
    f"{top_segment} customers "
    f"generate the highest revenue."
)

print(
    f"This segment contributes "
    f"{segment_share:.2f}% "
    f"of total business revenue."
)

print(
    "Customer retention strategies "
    "should prioritize this segment."
)
```

AUTOMATED BUSINESS INSIGHT

New customers generate the highest revenue.
 This segment contributes 20.62% of total business revenue.
 Customer retention strategies should prioritize this segment.

Quantifiable Recommendation

Strategic Action

Increase retention among high-value customer segments through:

- Personalized offers

- VIP rewards
- Loyalty campaigns
- Personalized product recommendations

Estimated Business Impact

Improving retention among top-performing customer segments by **6%** may significantly increase long-term revenue.

```
In [39]: # =====  
# QUANTIFIABLE BUSINESS IMPACT  
# =====  
  
customer_retention_gain = (  
    total_revenue * 0.06  
)  
  
print(  
    f"Estimated Revenue Opportunity: "  
    f"${customer_retention_gain:,.2f}"  
)
```

Estimated Revenue Opportunity: \$1,788,226.11

Regional Marketing Performance

Business Objective

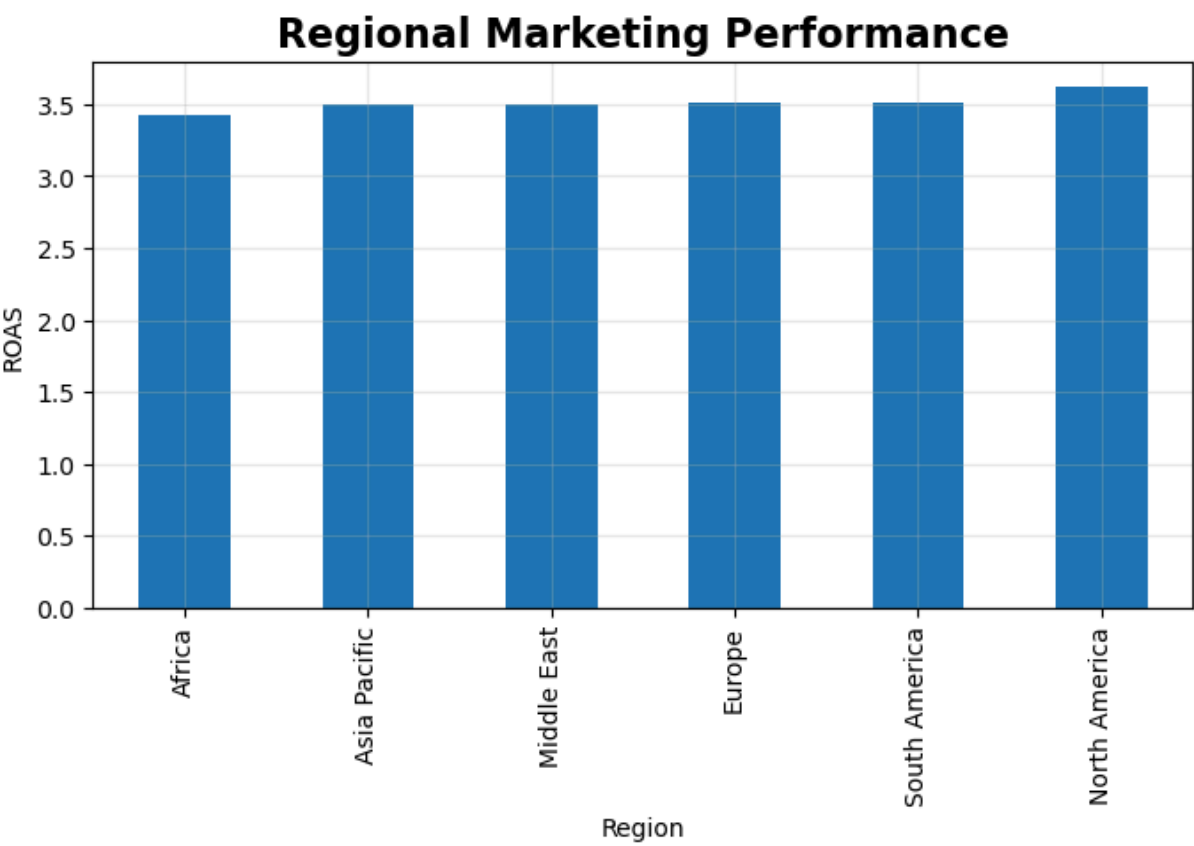
This analysis evaluates marketing performance across geographic regions.

The objective is to identify:

- High-performing regions
- Marketing efficiency differences
- Geographic revenue opportunities
- Regional expansion priorities

Understanding regional performance helps optimize marketing investment and geographic strategy.

```
In [41]: # =====  
# REGIONAL MARKETING PERFORMANCE  
# =====  
  
regional_performance = (  
    df.groupby("Region")  
    .agg({  
        "Revenue": "sum",  
        "MarketingCost": "sum"  
    })  
)  
  
regional_performance["ROAS"] = (  
    regional_performance["Revenue"]  
    /  
    regional_performance["MarketingCost"]  
)  
  
plt.figure(figsize=(8,4))  
  
regional_performance["ROAS"] \  
    .sort_values() \  
    .plot(kind="bar")  
  
plt.title(  
    "Regional Marketing Performance",  
    fontsize=16,  
    fontweight="bold"  
)  
  
plt.ylabel("ROAS")  
plt.xlabel("Region")  
  
plt.grid(alpha=0.3)  
  
plt.show()
```



```
In [42]: # =====
# AUTOMATED BUSINESS INSIGHT
# =====

best_region = (
    regional_performance["ROAS"]
    .idxmax()
)

best_region_roas = (
    regional_performance["ROAS"]
    .max()
)

print("AUTOMATED BUSINESS INSIGHT")
print("-" * 50)

print(
    f"{best_region} produces "
    f"the strongest regional "
    f"marketing efficiency at "
    f"{best_region_roas:.2f}x ROAS."
)

print(
    "This region may represent "
    "an opportunity for scaling "
    "marketing investment."
)
```

AUTOMATED BUSINESS INSIGHT

North America produces the strongest regional marketing efficiency at 3.62x ROA S.

This region may represent an opportunity for scaling marketing investment.

Quantifiable Recommendation

Strategic Action

Increase investment in high-performing regions while improving underperforming regional campaigns through:

- Regional campaign personalization
- Geo-targeted promotions
- Localized offers

Estimated Business Impact

Scaling successful regional strategies could improve revenue performance by approximately **5%**.

```
In [43]: # =====  
# QUANTIFIABLE BUSINESS IMPACT  
# =====  
  
regional_growth = (  
    total_revenue * 0.05  
)  
  
print(  
    f"Estimated Revenue Opportunity: "  
    f"${regional_growth:,.2f}"  
)
```

Estimated Revenue Opportunity: \$1,490,188.43

Executive Findings

Key Business Findings

1. Revenue Growth Opportunities Exist

Revenue trend analysis identified periods of stronger performance, suggesting opportunities for campaign optimization and seasonal planning.

2. Funnel Leakage is a Major Revenue Risk

A significant proportion of customers exit the funnel between cart creation and purchase completion.

3. Marketing Efficiency Varies by Channel

Certain acquisition channels produce significantly stronger marketing returns, creating opportunities for budget reallocation.

4. Device Performance Impacts Conversion

Lower-performing devices may be creating friction during checkout.

5. Customer Segments Contribute Unevenly

A small number of high-value customer segments contribute a disproportionate share of revenue.

6. Regional Opportunities Exist

Marketing efficiency differs significantly across regions, indicating geographic optimization potential.

Strategic Recommendations

Recommended Actions

1. Reduce Cart Abandonment

Optimize checkout flow and launch abandoned-cart campaigns.

Estimated Revenue Opportunity: +12%

2. Reallocate Marketing Spend

Shift budget toward highest-performing ROAS channels.

Estimated Revenue Opportunity: +7%

3. Improve Low-Converting Devices

Enhance UX and checkout performance.

Estimated Revenue Opportunity: +4%

4. Increase High-Value Customer Retention

Strengthen loyalty and personalized marketing.

Estimated Revenue Opportunity: +6%

5. Expand High-Performing Regional Strategies

Increase investment in strong-performing regions.

Estimated Revenue Opportunity: +5%

Combined Potential Business Impact

If successfully implemented, these initiatives may materially improve:

- Revenue growth
- Conversion performance
- Marketing efficiency
- Customer retention
- Customer lifetime value

```
In [44]: # =====
# TOTAL ESTIMATED BUSINESS IMPACT
# =====

total_business_impact = (
    revenue_recovery
    +
    marketing_gain
    +
    device_revenue_uplift
    +
    customer_retention_gain
    +
    regional_growth
)

print("="*60)
print("ESTIMATED TOTAL BUSINESS IMPACT")
print("="*60)

print(
    f"Estimated Revenue Opportunity: "
    f"${total_business_impact:,.2f}"
)
```

```
=====
ESTIMATED TOTAL BUSINESS IMPACT
=====
Estimated Revenue Opportunity: $10,133,281.31
```

Executive Conclusion

This consulting engagement demonstrates how customer behavior analytics and marketing intelligence can uncover measurable opportunities for business growth.

The analysis identified opportunities to:

- Improve customer conversion
- Reduce cart abandonment
- Optimize marketing efficiency
- Improve device experience
- Increase customer retention
- Expand successful regional strategies

By implementing the recommendations outlined in this report, leadership can strengthen revenue performance, improve marketing ROI, and build a more scalable eCommerce growth strategy.

In []:

